

Staphylococcus saprophyticus as a common cause of urinary tract infections.

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Until the last decade, coagulase-negative staphylococci occurring in urine specimens were usually regarded as a contaminant. In the early 1970s, i.e., more than ten years after the original demonstration of *Staphylococcus saprophyticus* in urine specimens, this species became recognized as a frequent cause of urinary tract infections (UTI). In young women, *S. saprophyticus* is, after *Escherichia coli*, the second-most-frequent causative agent of acute UTI. Patients with UTI caused by *S. saprophyticus* usually present with symptomatic cystitis. Signs and symptoms of renal involvement are also often registered. The urine sediment of a patient with UTI caused by *S. saprophyticus* has a characteristic appearance microscopically. Chemical screening methods for bacteriuria do not always succeed in diagnosing UTI caused by *S. saprophyticus*. Even when such an infection occurs above the neck of the bladder, low numbers of colony-forming units (less than 10⁵ cfu/ml) of *S. saprophyticus* are comparatively often found in the bladder and voided urine. *S. saprophyticus* is usually susceptible to antibiotics commonly prescribed for patients with UTI, with the exception of nalidixic acid. The bacterium has a capacity for selective adherence to human urothelium. It causes direct hemagglutination. The adhesin for *S. saprophyticus* is a lactosamine structure. This staphylococcal species produces an extracellular enzyme complex that can inhibit growth of both gram-positive and gram-negative bacteria.

Microbial virulence determinants and the pathogenesis of urinary tract infection.

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The most frequent and best-studied agent of urinary tract infection (UTI) is *Escherichia coli*, which serves as a useful model pathogen for understanding microbial virulence in relation to UTI pathogenesis. The *E. coli* strains that cause most UTIs and other extraintestinal *E. coli* infections represent a highly specialized subset of the total *E. coli* population. The enhanced virulence potential of such strains, which collectively are known as uropathogenic *E. coli* or extraintestinal pathogenic *E. coli* (ExPEC), is thought to be caused mainly by their multiple virulence factors. These virulence factors include diverse adhesins, siderophores, toxins, polysaccharide coatings, and other properties that assist the bacteria in avoiding or subverting host defenses, injuring or invading host cells and tissues, and stimulating a noxious inflammatory response. Although the true evolutionary basis for ExPEC is unknown, the virulence factors of ExPEC serve as useful epidemiologic markers and in the future may provide effective targets for anti-UTI interventions.

Host-pathogen interactions in urinary tract infection.

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The urinary tract is a common site of bacterial infections; nearly half of all women experience at least one urinary tract infection (UTI) during their lifetime. These infections are classified based on the condition of the host. Uncomplicated infections affect otherwise healthy individuals and are most commonly caused by uropathogenic *Escherichia coli*, whereas complicated infections affect patients with underlying difficulties, such as a urinary tract abnormality or catheterization, and are commonly caused by species such as *Proteus mirabilis*. Virulence and fitness factors produced by both pathogens include

fimbriae, toxins, flagella, iron acquisition systems, and proteins that function in immune evasion. Additional factors that contribute to infection include the formation of intracellular bacterial communities by *E. coli* and the production of urease by *P. mirabilis*, which can result in urinary stone formation. Innate immune responses are induced or mediated by pattern recognition receptors, antimicrobial peptides, and neutrophils. The adaptive immune response to UTI is less well understood. Host factors TLR4 and CXCR1 are implicated in disease outcome and susceptibility, respectively. Low levels of TLR4 are associated with asymptomatic bacteriuria while low levels of CXCR1 are associated with increased incidence of acute pyelonephritis. Current research is focused on the identification of additional virulence factors and therapeutic or prophylactic targets that might be used in the generation of vaccines against both uropathogens.

Genomic epidemiology and antibiotic susceptibility profiling of uropathogenic *Escherichia coli* among children in the United States.

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Escherichia coli is the most common cause of urinary tract infections (UTIs) in children, and yet the underlying mechanisms of virulence and antibiotic resistance and the overall population structure of the species is poorly understood within this age group. To investigate whether uropathogenic *E. coli* (UPEC) from children who developed pyelonephritis carried specific genetic markers, we generated whole-genome sequence data for 96 isolates from children with UTIs. This included 57 isolates from children with either radiologically confirmed pyelonephritis or cystitis and 27 isolates belonging to the well-known multidrug-resistant sequence type ST131, selected to investigate their population structure and antibiotic resistance characteristics. We observed a UPEC population structure that is similar to those reported in adults. In comparison with prior investigations, we found that the full *pap* operon was more common among UPEC from pediatric cases of pyelonephritis. Further, in contrast with recent reports that the P-fimbriae adhesin-encoding *papGII* allele is substantially more prevalent in invasive UPEC from adults, we found *papGII* was common to both invasive and non-invasive UPEC from children. Among the set of ST131 isolates from children with UTIs, we found antibiotic resistance was correlated with known genetic markers of resistance, as in adults. Unexpectedly, we observed that *fimH30*, an allele of the fimbrial gene *fimH* often used as a proxy to type ST131 isolates into the most drug-resistant subclade C, was carried by some of the subclade A and subclade B isolates, suggesting that the *fimH30* allele could confer a selective advantage for UPEC. **IMPORTANCE** Urinary tract infections (UTIs), which are most often caused by *Escherichia coli*, are not well studied in children. Here, we examine genetic characteristics that differentiate UTI-causing bacteria in children that either remain localized to the bladder or are involved in more serious kidney infections. We also examine patterns of antibiotic resistance among strains from children that are part of *E. coli* sequence type 131, a group of bacteria that commonly cause UTIs and are known to have high levels of drug resistance. This work provides new insight into the virulence and antibiotic resistance characteristics of the bacteria that cause UTIs in children.

Urinary tract infections.

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Urinary tract infection in women has its origin, predominantly, via ascending bacteria from the

periurethral microflora. Asymptomatic bacteriuria, except for the pregnant patient, need not be treated. *E. coli* is the most common bacterium to cause UTIs, and is usually susceptible to oral antibiotics. Patients who are hospitalized with an indwelling Foley catheter or who have undergone instrumentation, tend to be infected with a bacterium other than *E. coli*. Patients with uncomplicated cystitis can effectively be treated with an oral antibiotic (Table 1) for 3 days. Patients who do not respond to empiric therapy have a recurrence within 2 weeks of treatment, or who have a recurrence within the first week after treatment, should have a pretreatment.

Bacterial etiologic agents in the pathogenesis of urinary tract infection.

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Most of the information available concerning virulence factors of uropathogens is based on studies of *Escherichia coli*, the commonest cause of urinary tract infections. Earlier studies revealed several phenotypic *Escherichia coli* virulence factors that influenced both the anatomic level and severity of urinary tract infection. Virulence factors included O antigen serotype, presence and quantity of K capsular polysaccharide, adherence to uroepithelial cells, resistance to serum bactericidal activity, hemolysin, and aerobactin production. The introduction of DNA hybridization methodology has provided a valuable tool for reevaluation of the epidemiology of *Escherichia coli* infection as well as an alternative to the conventional phenotypic approach for studying the genotypic basis for virulence.

Uropathogenic bacteria leave a mark.

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Urinary tract infections are common, obnoxious and in some cases even life endangering. They are most often caused by uropathogenic *Escherichia coli*. Persistent and seemingly recurrent infections may be caused by bacteria establishing intracellular reservoirs in the urothelial epithelium. A better understanding of the mechanisms involved in the complex interplay between host and pathogens is required for preventing and controlling these infections. A paper in this issue of Laboratory Investigation reports changes in gene expression in urothelial carcinoma cells following infection by uropathogenic bacteria which go along with the activation of DNA methyltransferase 1 and increased methylation at the CDKN2A gene. These pioneer findings should stimulate the further development of in vitro models for studying urothelial infections and prompt more extensive analyses of epigenetic alterations elicited by bacterial infections in the urinary tract.

Urinary tract infection at a specialist hospital in Saudi Arabia.

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Midstream specimens of urine from inpatients and out patients at King Fahd Specialist Hospital in Buraidah, Saudi Arabia, were collected over a period of 12 months to determine prospectively the frequencies of different causative organisms and their antimicrobial susceptibility. A total of 854 from 4157 specimens (20.54%) gave significant bacterial counts i.e., counts greater than 10(5) organisms per

millilitre. *Escherichia coli* was the commonest organism (50.11%) followed by *Klebsiella* spp. (28.33%) *Pseudomonas* spp. (7.84%) and *Proteus* spp. (4.91%). Other bacterial pathogens were *Enterococcus* spp. (3.98%), *Acinetobacter* spp. (1.84%), *Staphylococcus aureus* (1.63%), *Enterobacter* spp. (0.35), *Staphylococcus epidermidis* (0.30%), *Haemolytic streptococci B* (0.47%) and *Salmonella paratyphi A* (0.12%). In vitro drug sensitivity tests showed norfloxacin and nalidixic acid to be very effective for most of the strains of the bacterial pathogens. A very high proportion of strains of *Escherichia coli* (86%) and *Klebsiella* spp. (94.6%) were found to be resistant to ampicillin.

Urinary tract infections: pathogenesis, host susceptibility and emerging therapeutics.

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Urinary tract infections (UTIs), which include any infection of the urethra, bladder or kidneys, account for an estimated 400 million infections and billions of dollars in health-care spending per year. The most common bacterium implicated in UTI is uropathogenic *Escherichia coli*, but diverse pathogens including *Klebsiella*, *Enterococcus*, *Pseudomonas*, *Staphylococcus* and even yeast such as *Candida* species can also cause UTIs. UTIs occur in both women and men and in both healthy and immunocompromised patients. However, certain patient factors predispose to disease: for example, female sex, history of prior UTI, or the presence of a urinary catheter or other urinary tract abnormality. The current clinical paradigm for the treatment of UTIs involves the use of antibiotics. Unfortunately, the efficacy of this approach is dwindling as the prevalence of antimicrobial resistance rises among UTI isolates, and the immense quantity of antibiotics prescribed annually for these infections contributes to the emergence of resistant pathogens. Therefore, there is an urgent need for new antibiotics and non-antibiotic treatment and prevention strategies. In this Review, we discuss how recent studies of bacterial pathogenesis, recurrence, persistence, host-pathogen interactions and host susceptibility factors have elucidated new and promising targets for the treatment and prevention of UTIs.

Causative agents and antimicrobial susceptibilities of urinary tract infections in the northwest of Iran.

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The empirical therapy of urinary tract infections (UTI) relies on the predictability of the agents causing UTI and knowledge of their antimicrobial susceptibility patterns. In a prospective study undertaken over a 14-month period, 5136 samples from patients suspected of having a UTI were analyzed, of which 676 were culture-positive. Isolated bacteria were identified by standard tests, and antibiotic susceptibility was determined by disk diffusion method. According to our results, *Escherichia coli* was the most common etiological agent of UTI (74.6%), followed by *Klebsiella* spp (11.7%), *Staphylococcus saprophyticus* (6.4%), and *Pseudomonas aeruginosa* (2.2%). Analysis of the frequency of isolated bacteria according to the age of the patients revealed that *Klebsiella* infections are more prevalent in the older age groups (>10 years) and *Pseudomonas* infections are more prevalent in children and the elderly (60 years). Results of antimicrobial susceptibility analysis for *E. coli*, as the most prevalent cause of UTI, to commonly used antibiotics are as follows: amikacin (97.8%), gentamicin (97%), ciprofloxacin (94%), nitrofurantoin (87.1%), nalidixic acid (93.7%), trimethoprim-sulfamethoxazole (48.2%), cephalexin (76%), and ampicillin (6.9%). The results show that the antimicrobial resistance patterns of the causes of UTI are

highly variable and continuous surveillance of trends in resistance patterns of uropathogens is important.
